

HALL π -SUBGROUPS AND CHARACTERS OF π' -DEGREE

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ABSTRACT. We study the relationship between the existence of Hall π -subgroups and that of irreducible characters of π' -degree with prescribed fields of values in finite groups. This work extends a result of Navarro and Tiep from a single odd prime to multiple odd primes.

1. INTRODUCTION

A well-known conjecture of R. Gow, proved by G. Navarro and P. H. Tiep [NT08], asserts that if G is a finite group of even order, then G possesses a nontrivial (complex) irreducible character of odd degree whose values are rational. This result was subsequently generalized to all primes by the same authors, as follows.

Theorem 1.1 ([NT06], Theorem A). *Let G be a finite group of order divisible by a prime p . Then G has a nontrivial irreducible character of degree not divisible by p whose values lie in $\mathbb{Q}(e^{2\pi i/p})$.*

In [GHSV21], together with A. A. Schaeffer Fry and C. Vallejo, we attempted to extend this theorem from a single prime p to a set π of two primes. Among other results, it was shown that if $\pi = \{2, p\}$ and $\gcd(|G|, 2p) > 1$, then G possesses a nontrivial irreducible character of π' -degree whose values are contained in $\mathbb{Q}(e^{2\pi i/p})$. (Here, a character χ is said to have π' -degree if $\chi(1)$ is not divisible by any prime in π .) Unfortunately, this phenomenon does not hold in general when π consists of two odd primes. For example, as pointed out in [GHSV21, Proposition 4.1], the Tits group ${}^2F_4(2)'$ has no nontrivial irreducible character of $\{3, 5\}'$ -degree with values in $\mathbb{Q}(e^{2\pi i/15})$.

In this paper, we propose a different extension – perhaps a more natural one – of Theorem 1.1 that works for an arbitrary set of odd primes. In the following, we write $\pi(G)$ for the set of prime divisors of $|G|$. Recall that a subgroup $H \leq G$ is called a Hall π -subgroup of G if $\pi(H) \subseteq \pi$ and $|G : H|$ is not divisible by any prime in π .

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Theorem A. *Let π be a set of odd primes and G a finite group possessing a nontrivial Hall π -subgroup. Then G has a nontrivial π' -degree irreducible character with values in $\mathbb{Q}(e^{2\pi i/p})$ for some $p \in \pi$.*

Theorem A may be viewed as a generalization of Theorem 1.1. In fact, when p is odd, Theorem 1.1 follows from the case $\pi = \{p\}$ of Theorem A together with the first Sylow theorem.

If we assume that $2 \in \pi$, then the conclusion of Theorem A remains true when $|\pi| \leq 2$, as noted above, but fails in general once $|\pi| \geq 3$. For example, if G is any non-abelian simple group and $\pi = \pi(G)$, then the trivial character 1_G is the only irreducible character of G with π' -degree. There are also counterexamples with $\pi \subsetneq \pi(G)$, such as $(G, \pi) = (A_7, \{2, 3, 5\})$. At present, we do not have a conceptual explanation for why the prime 2 is special in this context.

Similar to Gow's conjecture and the results in [NT06, GHSV21], Theorem A admits a clean reduction to finite simple groups. Accordingly, the main body of this paper is devoted to proving the result for such groups. This requires some new work on the relationship between the existence of Hall π -subgroups and π' -degree irreducible characters in simple groups of Lie type.

2. NON-ABELIAN SIMPLE GROUPS

This section is devoted to proving Theorem A for all non-abelian simple groups. We first fix some notation. For a positive integer k , we let $\zeta_k := e^{2\pi i/k}$ and we write $\mathbb{Q}(\zeta_k)$ for the k th cyclotomic field. As usual, $\text{Irr}(G)$ denotes the set of all irreducible characters of a group G , and $\text{Irr}_{\pi'}(G)$ denotes the subset consisting of those characters whose degrees are not divisible by any prime in π . For $\chi \in \text{Irr}(G)$, we write $\mathbb{Q}(\chi)$ for the field of values of χ , that is, the smallest extension of $\mathbb{Q}$ containing all values of χ . Finally, for integers $x \leq y$, we let $[x, y] := \{z \in \mathbb{Z} \mid x \leq z \leq y\}$.

As mentioned above, we aim at proving the following.

Theorem 2.1. *Let S be a non-abelian simple group and π be a set of odd primes such that S has a Hall π -subgroup. Then there exists $1_S \neq \chi \in \text{Irr}_{\pi'}(S)$ such that $\mathbb{Q}(\chi) \subseteq \mathbb{Q}(\zeta_p)$ for some $p \in \pi$.*

Hall π -subgroups for a set π of odd primes are relatively common in simple groups of Lie type, but are much more restricted in alternating and sporadic groups. We begin by treating the alternating and sporadic cases.

Lemma 2.2. *Let $n \in \mathbb{N}_{\geq 5}$ and let $\pi \subseteq [1, n]$ be a set of prime numbers. Assume that $2 \notin \pi$. Then the alternating group A_n admits a Hall π -subgroup if and only if $|\pi| = 1$.*

Proof. If $|\pi| = 1$ the statement is obviously implied by the Sylow theory. For the other implication, assume that A_n admits a Hall π -subgroup H and suppose for a contradiction that $|\pi| \geq 2$. In particular let $\pi = \{p_1, p_2, \dots, p_t\}$ for some odd primes

$p_1 < p_2 < \cdots < p_t$. Notice that H is solvable, by the Feit–Thompson theorem [FT63]. By [Hal28] we then know that H admits a $\{p_1, p_2\}$ -Hall subgroup K . It follows that K is a solvable $\{p_1, p_2\}$ -Hall subgroup of A_n . Moreover, since $2 \notin \{p_1, p_2\}$ we also have that K is a solvable $\{p_1, p_2\}$ -Hall subgroup of the symmetric group S_n . Using [Hal56, Theorem A4], we deduce that $|K|$ is even, and therefore $2 \in \{p_1, p_2\} \subseteq \pi$. This clearly contradicts our hypothesis. $\square$

We note that the use of the Feit–Thompson theorem can be avoided by instead relying on the known classification of non-solvable Hall subgroups of symmetric groups. Suppose that H is a non-solvable Hall π -subgroup of A_n . Since $2 \notin \pi$, it follows that H is also a non-solvable Hall π -subgroup of S_n . This family of subgroups of symmetric groups is completely described in [Tho66]. In particular, either $H = S_n$ or n is a prime number and $H = S_{n-1}$. In both cases, we would have $2 \in \pi$, which again contradicts our hypothesis.

Proposition 2.3. *Theorem 2.1 holds when S is an alternating group, a sporadic group, or the Tits group ${}^2F_4(2)'$.*

Proof. The case of alternating groups follows from Lemma 2.2 and Theorem 1.1. We now consider the sporadic groups and the Tits group. By [Gro86, Theorem 6.14], if S has a Hall π -subgroup for a set π of odd primes, then $|\pi| \leq 2$. The result then follows from [GHSV21, Theorem 2.1], except possibly in the cases $(S, \pi) = ({}^2F_4(2)', \{3, 5\})$, $(J_4, \{23, 43\})$, or $(J_4, \{29, 43\})$. However, a direct check using [Atl] shows that in each of these cases, S does not have a Hall π -subgroup. $\square$

Remark 2.4. The hypothesis $2 \notin \pi$ in Proposition 2.3 can not be removed. Indeed, as already pointed out in the introduction, A_7 possesses a Hall $\{2, 3, 5\}$ -subgroup but the only $\{2, 3, 5\}$ -degree irreducible character of A_7 is the trivial character. This observation can be extended to any prime larger than 7. Let $p \geq 11$ be a fixed prime number, and let π be the set consisting of all prime numbers strictly smaller than p . In this case A_{p-1} is a Hall π -subgroup of A_p but $\text{Irr}_{\pi'}(A_p) = \{1_{A_p}\}$. In fact any non-trivial irreducible character χ of π' -degree of A_p would satisfy $\chi(1) = p$. This is not possible because for any $\zeta \in \text{Irr}(S_p)$ we have either $\zeta(1) \leq p - 1$ or $\zeta(1) \geq \frac{p(p-3)}{2}$, by [Ras77]. Since $p - 3 > 4$ we have that for any $\theta \in \text{Irr}(A_p)$, either $\theta(1) \leq p - 1$ or $\theta(1) > p$.

We now turn to the proof of Theorem 2.1 for simple groups of Lie type $S \neq {}^2F_4(2)'$. By this, we mean simple groups of the form $S = G/\mathbf{Z}(G)$, where $G := \mathbf{G}^F$ is the group of fixed points of a simple, simply connected algebraic group $\mathbf{G}$ defined over an algebraically closed field of characteristic ℓ , under a Steinberg endomorphism F of $\mathbf{G}$.

The case $\ell \notin \pi$ is easy.

Proposition 2.5. *Theorem 2.1 holds when S is a simple group of Lie type in characteristic ℓ and $\ell \notin \pi$.*

Proof. The Steinberg character St_G of G is trivial on $\mathbf{Z}(G)$ and thus can be viewed as a character of S . Moreover, St_G is rational-valued and has degree a power of ℓ ; see [GM20, Proposition 3.4.10]. $\square$

We therefore focus on the case $\ell \in \pi$. Let $\mathbf{B}$ be an F -stable Borel subgroup of $\mathbf{G}$ and $\mathbf{T}$ a (maximally-split) F -stable maximal torus of $\mathbf{G}$ inside $\mathbf{B}$. Let Φ be the root system of $\mathbf{G}$ with respect to $\mathbf{T}$ and $\mathbf{B}$ and Φ^+ the set of corresponding positive roots. Let $\mathbf{U}$ be the product of the root subgroups corresponding to the roots in Φ^+ . This $\mathbf{U}$ is indeed the unipotent radical of $\mathbf{B}$. We have $U := \mathbf{U}^F \in \text{Syl}_\ell(G)$ and $B = UT$, where $U := \mathbf{U}^F$ and $T := \mathbf{T}^F$; see [Car85, §1.9 and §2.9].

Let $(\mathbf{G}^*, F^*)$ be the pair dual to $(\mathbf{G}, F)$, and set $G^* := \mathbf{G}^{*F^*}$. Note that $S = [G^*, G^*]$, as S is simple. Let $\mathbf{T}^*$ be an F^* -stable maximal torus of $\mathbf{G}^*$ that is dual to $\mathbf{T}$ in the sense of [Car85, Proposition 4.3.1]. Let $\mathbf{B}^*$ be an F^* -stable Borel subgroup of $\mathbf{G}^*$ containing $\mathbf{T}^*$. Write $T^* := \mathbf{T}^{*F^*}$ and $B^* := \mathbf{B}^{*F^*}$. If $\mathbf{U}^*$ denotes the unipotent radical of $\mathbf{B}^*$ and $U^* := \mathbf{U}^{*F^*}$, then $B^* = U^*T^*$.

Lemma 2.6. *Let S be a simple group of Lie type in characteristic ℓ . Suppose that π is a set of odd primes containing ℓ (in particular, ℓ is odd) and that S has a Hall π -subgroup. Then B^* contains a Hall π -subgroup of G^* .*

Proof. Note that $\mathbf{Z}(G) \leq T \leq B$. By [Gro86, Theorem 3.2], the quotient $B/\mathbf{Z}(G)$, which may be regarded as a Borel subgroup of S , contains a Hall π -subgroup of S . Equivalently, $|G : B| = |S : B/\mathbf{Z}(G)|$ is a π' -number. By Corollary 4.4.2 and Proposition 4.4.4 of [Car85], we have $|G^*| = |G|$ and $|T^*| = |T|$, and it follows that $|U^*| = |U|$ and $|B| = |B^*|$. (Note that U and U^* have orders equal to the ℓ -parts of $|G|$ and $|G^*|$, respectively.) Now we have that $|G^* : B^*|$ is a π' -number. As B^* is solvable, it therefore contains a Hall π -subgroup of G^* . $\square$

Lemma 2.7. *Let $S \neq {}^2F_4(2)'$ be a simple group of Lie type in characteristic ℓ . To prove Theorem 2.1 for S , it is sufficient to assume that $\pi \cap \pi(T^*) \neq \emptyset$.*

Proof. By Proposition 2.5, we may assume that $\ell \in \pi$. If $\pi \cap \pi(S) = \{\ell\}$, then Theorem 2.1 follows from Theorem 1.1. We therefore may assume that $|\pi \cap \pi(S)| \geq 2$. Since S has a Hall π -subgroup, by Lemma 2.6, the Borel subgroup B^* contains a Hall π -subgroup of G^* . In particular, $|\pi \cap \pi(B^*)| \geq 2$, and hence there exists at least one prime lying in both π and $\pi(T^*)$. $\square$

Recall that $S = G/\mathbf{Z}(G)$, so the irreducible characters of S are precisely those irreducible characters of G whose kernel contains $\mathbf{Z}(G)$. The set $\text{Irr}(G)$ admits a natural partition into Lusztig series $\mathcal{E}(G, s)$ indexed by G^* -conjugacy classes of semisimple elements $s \in G^*$. The series $\mathcal{E}(G, s)$ consists of those irreducible characters of G that are constituents of some Deligne–Lusztig character $R_{\mathbf{S}}^{\mathbf{G}}(\theta)$, where $\mathbf{S}$ is an F -stable

maximal torus of $\mathbf{G}$ and $\theta \in \text{Irr}(\mathbf{S}^F)$ is such that the geometric conjugacy class of the pair $(\mathbf{S}, \theta)$ corresponds to the G^* -conjugacy class containing s . See [GM20, §2.6].

We will look for the desired character among the so-called *semisimple characters*. Recall that $\rho \in \text{Irr}(G)$ is called a semisimple character if the average value of ρ on $\mathfrak{C}^F$ is nonzero, where $\mathfrak{C}$ is the conjugacy class of $\mathbf{G}$ consisting of regular unipotent elements. For each semisimple element $s \in G^*$, the Lusztig series $\mathcal{E}(G, s)$ contains one or more semisimple characters, all of which have degree

$$|G^* : \mathbf{C}_{G^*}(s)|_{\ell'} \uparrow$$

In fact, when $\mathbf{C}_{G^*}(s)$ is connected, $\mathcal{E}(G, s)$ contains exactly one semisimple character; see [GM20, p. 171]. We record a well-known fact about these characters below. Here we use $\text{ord}(g)$ to denote the order of a group element g .

Lemma 2.8. *Assume the above notation. Assume furthermore that $\mathbf{C}_{G^*}(s)$ is connected. Then the unique semisimple character, say χ_s , in the series $\mathcal{E}(G, s)$ is trivial on $\mathbf{Z}(G)$ if and only if $s \in S$. In such situation, $\mathbb{Q}(\chi_s) \subseteq \mathbb{Q}(\zeta_{\text{ord}(s)})$.*

Proof. The first part follows from, for instance, [MT11, Proposition 24.21], [Mal07, Lemma 2.2], and [Hun24, Lemma 5.8]. The latter part is [GHSV21, Lemma 4.2]. $\square$

We handle the linear and unitary groups separately. To unify notation, we write $S = \text{PSL}_n^\epsilon(q)$, where the superscript $\epsilon = +1$ corresponds to the linear groups and $\epsilon = -1$ to the unitary groups. We use analogous notation for the related groups, for example $G = \text{SL}_n^\epsilon(q)$ and $G^* = \text{PGL}_n^\epsilon(q)$. It is more convenient to first study the characters of $\tilde{G} := \text{GL}_n^\epsilon(q)$ and then analyze those of $G = \text{SL}_n^\epsilon(q)$ as irreducible constituents of restricted characters. Note that $\tilde{G}$ is self-dual, and we will identify $\tilde{G}$ with its dual group. Let $\pi : \tilde{G} \rightarrow G^*$ be the natural projection from $\tilde{G}$ to G^* .

Let $\tilde{s}$ be a semisimple element of $\tilde{G}$. Since the ambient algebraic group of $\tilde{G}$ has connected center, the Lusztig series $\mathcal{E}(\tilde{G}, \tilde{s})$ contains a unique semisimple character, which we denote by $\chi_{\tilde{s}}$.

Lemma 2.9. *The number of irreducible constituents of the restriction of semisimple character $\chi_{\tilde{s}}$ to G divides $\gcd(\text{ord}(\tilde{s}), n, q - \epsilon)$.*

Proof. Set $s := \pi(\tilde{s})$. Recall that $\chi_{\tilde{s}}(1) = |\tilde{G} : \mathbf{C}_{\tilde{G}}(\tilde{s})|_{\ell'}$. The restriction of $\chi_{\tilde{s}}$ from $\tilde{G}$ to G is multiplicity-free and its irreducible constituents are precisely the semisimple characters of the Lusztig series $\mathcal{E}(G, s)$ by [GM20, Corollary 2.6.18]. Each of these constituent has degree $|G^* : \mathbf{C}_{G^*}(s)|_{\ell'}$. Therefore, the number of irreducible constituents of the restriction is

$$\frac{|\tilde{G} : \mathbf{C}_{\tilde{G}}(\tilde{s})|_{\ell'}}{|G^* : \mathbf{C}_{G^*}(s)|_{\ell'} \uparrow} = \frac{|\tilde{G} : \mathbf{C}_{\tilde{G}}(\tilde{s})|_{\ell'}}{|\tilde{G} : \pi^{-1}(\mathbf{C}_{G^*}(s))|_{\ell'} \uparrow} = \frac{|\pi^{-1}(\mathbf{C}_{G^*}(s)) : \mathbf{C}_{\tilde{G}}(\tilde{s})|_{\ell'} \uparrow}{\uparrow}$$

Let Z be the (cyclic) subgroup of the multiplicative group $\mathbb{F}_{q^2-1}^\times$ of order $q - \epsilon$. Consider the mapping $\kappa : \pi^{-1}(\mathbf{C}_{G^*}(s)) \rightarrow Z$ defined by $g\tilde{s}g^{-1} = \kappa(g)\tilde{s}$. This is indeed a homomorphism with $\text{Ker}(\kappa) = \mathbf{C}_{\tilde{G}}(\tilde{s})$. It follows that the index $|\pi^{-1}(\mathbf{C}_{G^*}(s)) : \mathbf{C}_{\tilde{G}}(\tilde{s})|$ is equal to the order of the image $\text{Im}(\kappa)$ of κ . Therefore, the lemma follows once we show that

$$|\text{Im}(\kappa)| \text{ divides } \gcd(\text{ord}(\tilde{s}), n, q - \epsilon).$$

First, it is clear that $|\text{Im}(\kappa)| \mid (q - \epsilon)$. Now let g be an arbitrary element of $\pi^{-1}(\mathbf{C}_{G^*}(s))$. We have $\det(\tilde{s}) = \det(g\tilde{s}g^{-1}) = \det(\kappa(g)\tilde{s}) = \kappa(g)^n \det(\tilde{s})$, and hence $\kappa(g)^n = 1$; equivalently, $\text{ord}(\kappa(g)) \mid n$. Finally, we observe that $\text{ord}(\tilde{s}) = \text{ord}(g\tilde{s}g^{-1}) = \text{ord}(\kappa(g)\tilde{s}) = \text{lcm}(\text{ord}(\kappa(g)), \text{ord}(\tilde{s}))$, which implies that $\text{ord}(\kappa(g)) \mid \text{ord}(\tilde{s})$. $\square$

The proof of Lemma 2.9 also yields the following well-known result; however, we only require the first statement.

Lemma 2.10. *Let $s := \pi(\tilde{s}) \in G^*$, where $\tilde{s}$ is a semisimple element of $\tilde{G}$. Suppose that the multiset of eigenvalues of $\tilde{s}$ is not invariant under multiplication by every nontrivial root of unity. Then the centralizer $\mathbf{C}_{G^*}(s)$ is connected. Moreover, if the multiset of eigenvalues of $\tilde{s}$ is not invariant under multiplication by every nontrivial element of the subgroup of $\mathbb{F}_{q^2-1}^\times$ of order $q - \epsilon$, then $\chi_{\tilde{s}}$ restricts irreducibly to G .*

Proof. For every $g \in \pi^{-1}(\mathbf{C}_{G^*}(s))$, we have $g\tilde{s}g^{-1} = \kappa(g)\tilde{s}$, and hence $g\tilde{s}g^{-1}$ and $\tilde{s}$ have the same eigenvalues. Therefore, if the multiset of eigenvalues of $\tilde{s}$ is not invariant under multiplication by every nontrivial element of the subgroup of $\mathbb{F}_{q^2-1}^\times$ of order $q - \epsilon$, the homomorphism κ must have trivial image, which proves the second statement.

A similar argument, now in the setting of algebraic groups $\tilde{\mathbf{G}} := \text{GL}_n^\epsilon(\overline{\mathbb{F}}_\ell)$ and $\mathbf{G}^* := \text{PGL}_n^\epsilon(\overline{\mathbb{F}}_\ell)$ instead, shows that if the multiset of eigenvalues of $\tilde{s}$ is not invariant under multiplication by every nontrivial root of unity, then $\mathbf{C}_{G^*}(s)$ is the image of the connected group $\mathbf{C}_{\tilde{\mathbf{G}}}(\tilde{s})$ under π and hence is itself connected. $\square$

Proposition 2.11. *Theorem 2.1 holds for the simple groups $S = \text{PSL}_n(q)$ and $\text{PSU}_n(q)$, where $n \geq 2$ and q is a prime power.*

Proof. Recall that T^* is a maximal torus contained in a Borel subgroup B^* of G^* . By Lemma 2.7, we may assume that π and $\pi(T^*)$ share a common prime, say p . In the linear case, we have

$$|T^*| = (q - 1)^{n-1}$$

so p divides $q - 1$. In the unitary case,

$$|T^*| = \begin{cases} (q^2 - 1)^{(n-1)/2}, & \text{if } n \text{ is odd,} \\ (q^2 - 1)^{(n-2)/2} (q - 1), & \text{if } n \text{ is even,} \end{cases}$$

and hence p divides $q^2 - 1$ (see [Car85, p. 74]). Since B^* contains a Hall π -subgroup of G^* by Lemma 2.6, a comparison of the orders of G^* and B^* shows that p cannot divide $q + 1$. Therefore, as in the linear case, we again conclude that p divides $q - 1$.

We first consider the case $\epsilon = 1$. Let δ be a generator of $\mathbb{F}_q^\times$, and define the semisimple element

$$\tilde{s} := \text{diag}(\delta^{(q-1)/p}, \delta^{(q-1)(p-1)/p}, 1^{n-2}) \in \tilde{G}.$$

Let s denote the image $\pi(\tilde{s})$ of $\tilde{s}$ under $\pi : \tilde{G} \rightarrow G^*$. Then

$$\text{ord}(s) = \text{ord}(\tilde{s}) = p.$$

Taking conjugation if necessary, we may assume that $s \in T^*$. Moreover, as $\det(\tilde{s}) = \delta^{q-1} = 1$, we have

$$\tilde{s} \in G \quad \text{and} \quad s \in S \cap T^*.$$

Note that the case $(n, p) = (3, 3)$ cannot occur. (Otherwise, by Lemma 2.6, the Borel subgroup B^* , of order $q^3(q-1)^2$, would contain a Sylow 3-subgroup of G^* whose order $q^3(q^2-1)(q^3-1)$ has larger 3-part, leading to a contradiction.) It is then easy to check that the multiset of eigenvalues of $\tilde{s}$ is not invariant under multiplication by every nontrivial root of unity. By Lemma 2.10, it follows that $\mathbf{C}_{G^*}(s)$ is connected, and we let χ_s be the unique semisimple character in the series $\mathcal{E}(G, s)$. Using Lemma 2.8, we deduce that

$$\mathbb{Q}(\chi_s) \subseteq \mathbb{Q}(\zeta_{\text{ord}(s)}) = \mathbb{Q}(\zeta_p).$$

Moreover, since $s \in T^*$ and T^* is abelian, we have

$$T^* \leq \mathbf{C}_{G^*}(s).$$

It follows that

$$|G^* : \mathbf{C}_{G^*}(s)| \text{ divides } |G^* : T^*|$$

and hence

$$\chi_s(1) = |G^* : \mathbf{C}_{G^*}(s)|_{\ell'} \text{ divides } |G^* : T^*|_{\ell'}.$$

As $|G^* : T^*|_{\ell'} = |G^* : B^*|$ is a π' -number by Lemma 2.6, we conclude that $\chi_s(1)$ is also a π' -number. Also, from the fact that $s \in S = [G^*, G^*]$, we know that χ_s is trivial on $\mathbf{Z}(G)$. This completes the proof for linear groups.

It remains to consider the case $\epsilon = -1$ and $p \mid (q-1)$. Let $\tilde{t}$ be a semisimple element of order p in a maximal torus of order $q^2 - 1$ of $\text{GU}_2(q)$. Since $p \mid (q-1)$ and p is odd, we have $\gcd(p, q+1) = 1$, and hence $\tilde{t} \in \text{SU}_2(q)$. Define the semisimple element

$$\tilde{s} := \text{diag}(\tilde{t}, I_{n-2}) \in \tilde{G},$$

and consider the corresponding semisimple character $\chi_{\tilde{s}} \in \text{Irr}(\tilde{G})$. Note that $\tilde{s}$ belongs to a maximal torus of $\tilde{G}$ of order $(q^2-1)^{(n-1)/2}(q+1)$ if n is odd, and of order $(q^2-1)^{n/2}$ if n is even.

Note also that $\text{ord}(\tilde{s}) = \text{ord}(\tilde{t}) = p$, which does not divide $q + 1$. Lemma 2.9 then implies that $\chi_{\tilde{s}}$ restricts irreducibly to G . Arguments similar to those in the linear case show that this restriction is trivial on $\mathbf{Z}(G)$, has π' -degree, and takes values in $\mathbb{Q}(\zeta_p)$. This completes the proof. $\square$

We can now finish the proof of Theorem 2.1.

Proof of Theorem 2.1. By Propositions 2.3 and 2.11, and the classification of finite simple groups, we may assume that $S \neq {}^2F_4(2)'$ is a simple group of Lie type not of type A. By Lemma 2.7, there exists at least one prime lying in both π and $\pi(T^*)$. As before, let p be such a prime.

Note that

$$|T^* : (T^* \cap S)| = |T^*S : S| \text{ divides } |G^* : S|$$

and $|G^* : S|$ is the order of the group of *diagonal automorphisms* of S . On the other hand, the order of T^* is given by

$$|T^*| = \prod_{\alpha \in \mathcal{O}} (q^{|\alpha|} - 1),$$

where $\mathcal{O}$ is the set of F^* -orbits on the simple roots of the root system of $\mathbf{G}^*$ associated with $\mathbf{B}^*$ and $\mathbf{T}^*$, and q is the absolute value of the eigenvalues of F^* on the character group of $\mathbf{T}^*$; see [Car85, p. 74]. A straightforward case-by-case check, using [Atl, Table 5] for the size of the group of diagonal automorphisms and [Car85, §1.19] for the sizes of the F^* -orbits, reveals that if an odd prime p is a divisor of $|G^* : S|$, then indeed $(|G^* : S|_p)^2$ divides $|T^*|$, and it follows that p also divides $|T^* \cap S|$. (The only exception is the case $S = \text{PSU}_3(q)$ with $(q + 1)_3 = 3$, but this was already excluded using Proposition 2.11.) Therefore, we may and will assume from now on that

$$p \in \pi \cap \pi(T^* \cap S).$$

Consequently, there exists a semisimple element $s \in G^*$ such that

$$s \in T^* \cap S \quad \text{and} \quad \text{ord}(s) = p.$$

Suppose first that $p \nmid |\mathbf{Z}(\mathbf{G})|$. Then $\mathbf{C}_{\mathbf{G}^*}(s)$ is connected by [B-S70, Corollary 4.6]. Arguing as in the proof of Proposition 2.11 and applying Lemmas 2.6 and 2.8, we have that the unique semisimple character χ_s in $\mathcal{E}(G, s)$ has π' -degree, is trivial on $\mathbf{Z}(G)$, and satisfies $\mathbb{Q}(\chi_s) \subseteq \mathbb{Q}(\zeta_p)$.

Next suppose that $p \mid |\mathbf{Z}(\mathbf{G})|$. By [GLS98, Theorem 1.12.5] and the assumption that p is odd, we are left with only the case $p = 3$ and $\mathbf{G}$ to be of type E_6 . Here, $G = E_6^\epsilon(q)$ with $\epsilon \in \{\pm 1\}$, where $\epsilon = 1$ corresponds to the untwisted groups and $\epsilon = -1$ to the twisted groups, and $|\mathbf{Z}(G)| = \gcd(3, q - \epsilon) = 3$.

As mentioned above, all the semisimple characters in the Lusztig series $\mathcal{E}(G, s)$ indexed by s have the same degree $|G^* : \mathbf{C}_{G^*}(s)|_\ell$. This is a π' -number due to the fact that $s \in T^*$, as argued in the proof of Proposition 2.11. Moreover, as noted in the proof of [GHSV21, Proposition 4.5], these semisimple characters take integer

values on unipotent elements and therefore have field of values contained in $\mathbb{Q}(\zeta_3)$, by [GHSV21, Lemma 4.3]. Finally, since their degrees are coprime to 3, their restrictions to $\mathbf{Z}(G)$ are multiples of the trivial character. (Let χ be such a character and let $\chi_{\mathbf{Z}(G)} = \chi(1)\alpha$ for some $\alpha \in \text{Irr}(\mathbf{Z}(G))$. Then $\mathbf{1}_{\mathbf{Z}(G)} = \det(\chi)_{\mathbf{Z}(G)} = \det(\chi_{\mathbf{Z}(G)}) = \det(\chi(1)\alpha) = \alpha^{\chi(1)}$, which implies that the order of α divides $(\chi(1), |\mathbf{Z}(G)|)$.) This completes the proof. $\square$

3. A PROOF OF THEOREM A

We are now in the position to give a complete proof of our main result, as stated in the introduction.

A finite group admitting a Hall π -subgroup is often called an E_π -group. There is a large literature on the theory of E_π -groups, including the results about simple groups we cited in the previous section. We refer the reader to [RV10] for the latest results and relevant information. Note that the class of E_π -groups is not closed under extensions and a subgroup of an E_π -group might be no longer an E_π -group. Nevertheless, the only fact we need is that every normal/subnormal subgroup and every quotient of an E_π -group is an E_π -group. In fact, if H is a Hall π -subgroup of G and $N \trianglelefteq G$, then $H \cap N$ is a Hall π -subgroup of N and HN/N is a Hall π -subgroup of G/N .

Another fact we need is that if $N \trianglelefteq G$ such that $|G : N| = r$ is an odd prime and $\theta \in \text{Irr}(N)$ with $\mathbb{Q}(\theta) \subseteq \mathbb{Q}(\zeta_p)$ for some prime $p \neq r$, then there exists an irreducible constituent χ of θ^G such that $\mathbb{Q}(\chi) \subseteq \mathbb{Q}(\zeta_p)$. This follows from [GHSV21, Lemma 2.2], for instance.

We restate Theorem A for the reader's convenience.

Theorem 3.1. *Let π be a set of odd primes and G a finite group such that G has a nontrivial Hall π -subgroup. Then G possesses a nontrivial π' -degree irreducible character with values in $\mathbb{Q}(\zeta_p)$ for some $p \in \pi$.*

Proof. We argue by induction on $|G|$. Let $M \triangleleft G$ be a normal subgroup such that G/M is simple.

First assume that G/M is abelian and that $|G/M| \in \{2\} \cup \pi$. Then the inflation to G of any nontrivial linear character of G/M satisfies the required conditions. Now suppose that $r := |G/M|$ is an odd prime not belonging to π . In particular, we have $\pi(M) \supseteq \pi \cap \pi(G)$.

By the induction hypothesis and the fact that M also has a Hall π -subgroup, there exists $\psi \in \text{Irr}_{\pi'}(M)$ and some $p \in \pi$ such that $\mathbb{Q}(\psi) \subseteq \mathbb{Q}(\zeta_p)$. Let $\chi \in \text{Irr}(G)$ lie over ψ . By [Isa06, Corollary 6.19], either $\chi_M = \psi$ or $\chi_M = \sum_{i=1}^r \psi_i$ is the sum of the G -conjugates of ψ .

In the latter case, we may take $\chi = \psi^G$, with a note that $\chi(1) = r\psi(1)$ is a π' -number and $\mathbb{Q}(\chi) \subseteq \mathbb{Q}(\psi) \subseteq \mathbb{Q}(\zeta_p)$. In the former case, every irreducible character of

G lying over ψ is an extension of ψ , and hence has π' -degree. Moreover, as mentioned above, one of these extensions has values in $\mathbb{Q}(\zeta_p)$, as required.

Finally, suppose that G/M is non-abelian. Set $\pi_1 := \pi(G/M) \cap \pi$. (Note that π_1 could be empty.) Again, the group G/M has a Hall π -subgroup, which is also a Hall π_1 -subgroup. Theorem 2.1 then yields a character $\chi \in \text{Irr}_{\pi_1'}(G/M)$ such that $\mathbb{Q}(\chi) \subseteq \mathbb{Q}(\zeta_p)$ for some $p \in \pi_1$. Since $\chi(1)$ divides $|G/M|$, the character χ is also of π' -degree. The desired character is obtained by inflating χ to G . $\square$

We conclude with a remark that, in view of the above proof, the conclusion of Theorem A remains valid when $2 \in \pi$, provided that the group G is π -separable.

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